Chapter 06: Conclusion And Future Scope

6.1 Conclusion

The smart waste detection segregation platform uses computer vision and artificial intelligence to classify the waste automatically. It also combines a snappy past a Pi backend with a refined mobile-net version 2 architecture. Our setup also includes an interactive React front end that handles real-time camera detection. The system's overall accuracy is 94%. Battery, biological, cardboard, clothing, glass, metal, paper, plastic, shoes, and trash are the 10 types of garbage that fall under the category.

The major contributions of the project include:

- Waste may be classified with high accuracy using a lightweight deep learning architecture that is appropriate for real-time applications.
- The real-time camera detection will eliminate the process of manual image uploading
- User-centric UI design, machine learning, backend API integration, and an end-to-end development workflow.
- Educational and Environmental Impact: We will provide the users with actionable sustainability information coupled with detection results.

Limitations:

- Overall, the system does pretty well. But along the way, it runs into some clear limitations. Performance does tend to drop off in low-light situations because computer vision faces difficulty in classification, it also performs a little bad when there's heavy motion involved with the camera detection.
- The model's accuracy really depends on the quality and diversity of the dataset it was built on. If we would increase the data inside the dataset then it would increase its performance. Currently our dataset might not include every kind of waste you'd come across out in the real world. World.
- Confidence scores can drop pretty fast as well. It happens especially if the background is cluttered. or when only half of the object is visible at the moment which makes it hard to classify for the model.
- Currently, it sticks to just 10 pre-defined classes which can be increased later on. Things outside of those can get misclassified without much trouble.

- We have not put in any edge-device optimizations yet. Nothing like TensorFlow Lite to make it work better. That is what keeps it from running smoothly on mobile hardware.
- Even though there are a few limitations, the platform has a strong base and has potential. It lays the groundwork for waste management solutions that can scale up and be practical in everyday use for common people as well.

6.2 Future Scope

1. Expansion of the dataset and waste categories

- We can add a lot more waste classes like e-waste, types of organic waste ,chemicals, medical waste and a bunch of other waste categories..

- We can also curate a larger, more diverse and real-world dataset which will not only improve accuracy but will also increase real world usage.

2. Deployment on Mobile and IoT Devices

- Convert the current model to TensorFlow Lite or ONNX.

- Deploying on:

    - Smartphone apps

    - Raspberry Pi

    - Smart dustbins and IoT waste stations

3. Improve Real-Time Camera Detection

- Optimize FPS performance.

- Add auto-tracking, object bounding boxes, and multi-object segmentation.

- Use YOLOv8 or EfficientDet for faster camera inference.

4. 3D Segregation and Robotic Arm Integration

- Combine classification :
    - Robotic arms for automated sorting
    - Conveyor belt trash systems
- Enable end-to-end automated waste segregation.

5. Edge-based Environmental Impact Calculator

- Proved more advanced metrices:
    - Lifecycle C02 emissions
    - Recycling savings
    - Biodegradability score
    - Environmental hazard level

6. Cloud Deployment and Multi-User Support

- Host the system on cloud servers (AWS/GCP).
- Enable user dashboards and the history of scanned waste items.
- Provide analytics to institutions, waste plants, or NGOs.

7. Segmentation Instead of Classification

- We will implement semantic segmentation to identify multiple waste objects in one frame.
- Useful for mixed waste in bins or public places.

8. Voice Feedback System

- Add audio infrastructure for recycling/disposal:
    - “This is plastic; send it to the recycling bin.”
    - “This item has a high carbon footprint.”